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Roll No

EX-7004(1)-CBGS

B.E. VII Semester

Examination, June 2020

Choice Based Grading System (CBGS)

High Voltage Engineering

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Discuss advantages of transmitting electrical power at high voltages. 7
b) Explain important applications of high voltage. 7
2. a) What do you understand by “intrinsic strength” of solid dielectric? How does breakdown occur due to electrons in a solid dielectric? 8
b) How do the temperature and moisture affect the breakdown strength of solid dielectrics? 6
3. a) Define Townsend’s first and second ionization coefficients. How is the condition for breakdown obtained in a Townsend discharge? 7
b) What is a Tesla coil? How are damped high frequency oscillations obtained from a Tesla coil? 7
4. a) Explain the different schemes for cascade connection of transformers for producing very high a.c. voltages. 7
b) Explain the different methods of producing switching impulses in test laboratories. 7

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5. a) Discuss the different methods of measuring high d.c voltages. What are the limitations in each method? 7
b) What is a mixed potential divider? How is it used for impulse voltage measurements? 7
6. a) Explain the principle and construction of an electrostatic voltmeter for very high voltages. What are its merits and demerits for high voltage a.c measurements? 8
b) What are the conditions to be satisfied by a potential divider to be used for impulse work? 6
7. a) Mention the different electrical tests done on isolators and circuit breakers. 7
b) Explain the method of impulse testing of high voltage transformers. What is the procedure adopted for locating the failure? 7
8. Write short notes on any two of the followings: 14
 - a) Cavity breakdown
 - b) Surge current measurement
 - c) Paschen's law and its significance.

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